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A Related work

Ferbach et al. (2024) observes that emerging theoretical analyses of self-consuming generative models (also known as iterative retraining) suggest model collapse or stability may occur depending on the proportion of generated data used in each retraining iteration. This does not contradict our theoretical result, as it addresses a fundamentally different experimental scenario. Upon examination, the cited stability result originates from Bertrand et al. (2024). While their framework demonstrates non-collapsing models, our hypothesized scenario leads to model collapse. The key distinction lies in the assumptions:

Bertrand et al. (2024) postulates a persistent clean dataset that remains uncontaminated by generated content. Subsequent models are trained on hybrid datasets combining this pristine source with outputs from prior models, examining how incremental additions of synthetic data affect model behavior. Their analysis shows stability can be maintained when synthetic data proportions remain limited.

Conversely, our study investigates a distinct paradigm where generated content progressively contaminates a single evolving dataset, as illustrated in Figure 1. In this irreversible contamination process, the original real-world data becomes asymptotically negligible through successive retraining cycles. Our research focuses specifically on this complete corpus pollution scenario, making it complementary to the work by Bertrand et al. (2024) while addressing different dimensions of the data contamination problem.

B Mathematical derivations

B.1 Proof of Proposition 2.3

Declare a real number S. Let

$$\forall v_i \quad \alpha_{\boldsymbol{x};v_i} \triangleq \hat{p}_1(v_i|\boldsymbol{x}) \cdot S + \hat{p}_1(v_i|\boldsymbol{x}) \cdot N_{\boldsymbol{x}} - N_{\boldsymbol{x};v_i},$$

then

$$S \geq \frac{N_{\boldsymbol{x};v_i} - \hat{p}_1(v_i|\boldsymbol{x}) \cdot N_{\boldsymbol{x}}}{\hat{p}_1(v_i|\boldsymbol{x})} \implies \alpha_{\boldsymbol{x};v_i} \geq 0,$$

$$S \geq \max_{v_i \in V} \frac{N_{\boldsymbol{x};v_i} - \hat{p}_1(v_i|\boldsymbol{x}) \cdot N_{\boldsymbol{x}}}{\hat{p}_1(v_i|\boldsymbol{x})} \implies \forall v_i \;\; \alpha_{\boldsymbol{x};v_i} \geq 0.$$

Notice that,

$$\begin{aligned}
\alpha_{\boldsymbol{x}} &= \sum_{v_i \in V} \alpha_{\boldsymbol{x};v_i} \\
&= \sum_{v_i \in V} \left(\hat{p}_1(v_i|\boldsymbol{x}) \cdot S + \hat{p}_1(v_i|\boldsymbol{x}) \cdot N_{\boldsymbol{x}} - N_{\boldsymbol{x};v_i}\right) \\
&= S \cdot \underbrace{\sum_{v_i \in V} \hat{p}_1(v_i|\boldsymbol{x})}_{=1} \; + \; N_{\boldsymbol{x}} \cdot \underbrace{\sum_{v_i \in V} \hat{p}_1(v_i|\boldsymbol{x})}_{=1} \; - \; \sum_{v_i \in V} N_{\boldsymbol{x};v_i} \\
&= S + N_{\boldsymbol{x}} - N_{\boldsymbol{x}} \quad (\because \sum_{v_i \in V} N_{\boldsymbol{x};v_i} = N_{\boldsymbol{x}}) \\
&= S.
\end{aligned}$$

Thus, $\forall v_i \; \alpha_{\boldsymbol{x};v_i} = \hat{p}_1(v_i|\boldsymbol{x}) \cdot \alpha_{\boldsymbol{x}} + \hat{p}_1(v_i|\boldsymbol{x}) \cdot N_{\boldsymbol{x}} - N_{\boldsymbol{x};v_i} \geq 0$, which further implies

$$\hat{p}_1(v_i|\boldsymbol{x}) = \frac{N_{\boldsymbol{x};v_i} + \alpha_{\boldsymbol{x};v_i}}{N_{\boldsymbol{x}} + \alpha_{\boldsymbol{x}}},$$

and $\alpha_{\boldsymbol{x};v_i}$ is non-negative.

Remark B.1. For any particular $\boldsymbol{x}$,

$$\alpha^{\star}_{\boldsymbol{x}} \triangleq \max_{v_i \in V} \frac{N_{\boldsymbol{x};v_i} - \hat{p}_1(v_i|\boldsymbol{x}) \cdot N_{\boldsymbol{x}}}{\hat{p}_1(v_i|\boldsymbol{x})},$$

is the minimum $\alpha_{\boldsymbol{x}}$ that satisfies the above proposition. $\alpha^{\star}_{\boldsymbol{x}} = 0$ if and only if the output distribution of the LM is the same as the distribution of the training corpus given $\boldsymbol{x}$, that is to say, there is no error, and

$$\forall v_i \quad \hat{p}_1(v_i|\boldsymbol{x}) = \frac{N_{\boldsymbol{x};v_i}}{N_{\boldsymbol{x}}}.$$

Furthermore, in this situation, $\forall v_i \; \alpha^{\star}_{\boldsymbol{x};v_i} = 0$. **This property further illustrates that we can use non-negative numbers to model the errors between the output distribution of an LM and the distribution of its training corpus.**

B.2 Proof of Proposition 2.4

The existence proof follows identical reasoning to Proposition 2.3 through parameter substitution:

$$\alpha_i[n] = \hat{p}_n(v_i|\boldsymbol{x}) \cdot S + \hat{p}_n(v_i|\boldsymbol{x}) \cdot y[n-1] - y_i[n-1],$$

where

$$S \geq \max_{v_i \in V} \frac{y_i[n-1] - \hat{p}_n(v_i|\boldsymbol{x}) \cdot y[n-1]}{\hat{p}_n(v_i|\boldsymbol{x})}$$

guarantees non-negativity.

B.3 Proof that Equation (5) is a solution to the recurrence Equation (4)

Substitute $n = 1$ into

$$\begin{cases} y_i[n] &= N_{\boldsymbol{x};v_i} + \sum_{j=1}^{n} \alpha_i[j] \\ y[n] &= N_{\boldsymbol{x}} + \sum_{j=1}^{n} \alpha[j] \end{cases},$$

we get

$$\begin{cases} y_i[1] &= N_{\boldsymbol{x};v_i} + \alpha_i[1] \\ y[1] &= N_{\boldsymbol{x}} + \alpha[1] \end{cases},$$

which satisfies the initial condition

$$\hat{p}_1(v_i|\boldsymbol{x}) = \frac{y_i[1]}{y[1]} = \frac{N_{\boldsymbol{x};v_i} + \alpha_i[1]}{N_{\boldsymbol{x}} + \alpha[1]}.$$

Substitute

$$\begin{cases} y_i[n-1] & = & N_{\boldsymbol{x};v_i} + \sum_{j=1}^{n-1} \alpha_i[j] \\ y[n-1] & = & N_{\boldsymbol{x}} + \sum_{j=1}^{n-1} \alpha[j] \end{cases}$$

into the right hand side of

$$\hat{p}_n(v_i|\boldsymbol{x}) = \frac{y_i[n-1] + \alpha_i[n]}{y[n-1] + \alpha[n]},$$

we get

$$\frac{y_i[n-1] + \alpha_i[n]}{y[n-1] + \alpha[n]} = \frac{N_{\boldsymbol{x};v_i} + \sum_{j=1}^{n-1} \alpha_i[j] + \alpha_i[n]}{N_{\boldsymbol{x}} + \sum_{j=1}^{n-1} \alpha[j] + \alpha[n]} = \frac{N_{\boldsymbol{x};v_i} + \sum_{j=1}^{n} \alpha_i[j]}{N_{\boldsymbol{x}} + \sum_{j=1}^{n} \alpha[j]} = \frac{y_i[n]}{y[n]} = \hat{p}_n(v_i|\boldsymbol{x}).$$

This shows that the obtained solution satisfies the original recurrence equation.

B.4 The thinking process of solving the recurrence Equation (6)

We conjecture that Equation (6) has a solution satisfying

$$y_i[n] = \frac{N_{\boldsymbol{x};v_i} + k\sum_{j=1}^{n-1} y_i[j]}{1 + (n-1)k} + \alpha_i[n] \tag{8}$$

Then $y_i[n-1]$ will satisfy

$$y_i[n-1] = \frac{N_{\boldsymbol{x};v_i} + k\sum_{j=1}^{n-2} y_i[j]}{1 + (n-2)k} + \alpha_i[n-1] \tag{9}$$

Multiply both sides of Equation (8) by $1 + (n-1)k$ and both sides of Equation (9) by $1 + (n-2)k$, we get

$$(1 + (n-1)k)y_i[n] = N_{\boldsymbol{x};v_i} + k\sum_{j=1}^{n-1} y_i[j] + (1 + (n-1)k)\alpha_i[n]$$

$$(1 + (n-2)k)y_i[n-1] = N_{\boldsymbol{x};v_i} + k\sum_{j=1}^{n-2} y_i[j] + (1 + (n-2)k)\alpha_i[n-1]$$

Subtract the two equations, we obtain

$$(1+(n-1)k)y_i[n] - (1+(n-2)k)y_i[n-1] = ky_i[n-1] + (1+(n-1)k)\alpha_i[n] - (1+(n-2)k)\alpha_i[n-1]$$

Add $(1 + (n-2)k)y_i[n-1]$ to both sides, we have

$$(1 + (n-1)k)y_i[n] = (1 + (n-1)k)y_i[n-1] + (1 + (n-1)k)\alpha_i[n] - (1 + (n-2)k)\alpha_i[n-1]$$

Divide both sides by $1 + (n-1)k$, we get

$$y_i[n] = y_i[n-1] + \alpha_i[n] - \frac{1 + (n-2)k}{1 + (n-1)k}\alpha_i[n-1]$$